# CHAPTER 3 — ELEMENTARY FUNCTIONS

## 1. The Exponential Function

### 1.1. Definition.

The exponential function $e^z$ is defined by

$$e^z = e^x \cos y + ie^x \sin y, \quad z = x + iy \in \mathbb{C},$$

where $e^x$ is the real exponential function. Using Euler's formula,

$$e^{iy} = \cos y + i \sin y,$$

We can write

$$e^z = e^x e^{iy}, \quad z = x + iy \in \mathbb{C}.$$

This is the polar representation of $e^z$, and thus,

$$|e^z| = e^x, \quad \arg e^z = y + 2n\pi, \quad n = 0, \pm 1, \pm 2, \ldots$$

Obviously,

$$e^{x+i0} = e^x, \quad e^{2\pi i} = 1, \quad e^{i\pi} = -1,$$

and

$$e^z \neq 0, \quad z \in \mathbb{C}.$$

### 1.2. Periodic.

Since $\cos y$ and $\sin y$ are $2\pi$ periodic functions, $e^z$ is a periodic function of $2\pi i$,

$$e^{z+2\pi i} = e^x \cos(y + 2\pi) + ie^x \sin(y + 2\pi) = e^x \cos y + ie^x \sin y = e^z, \quad z \in \mathbb{C}.$$

### 1.3. Addition Formula.

$$e^{z_1+z_2} = e^{z_1} e^{z_2}, \quad z_1, z_2 \in \mathbb{C}.$$

Let $z_1 = x_1 + iy_1$, $z_2 = x_2 + iy_2$. Then $z_1 + z_2 = x_1 + x_2 + i(y_1 + y_2)$,

$$e^{z_1+z_2} = e^{x_1+x_2} e^{i(y_1+y_2)},$$

and

$$e^{z_1} e^{z_2} = e^{x_1} e^{iy_1} e^{x_2} e^{iy_2} = e^{x_1+x_2} e^{iy_1} e^{iy_2}.$$

$$\begin{aligned} e^{iy_1}e^{iy_2} &= (\cos y_1 + i\sin y_1)(\cos y_2 + i\sin y_2) \\ &= \cos y_1\cos y_2 - \sin y_1\sin y_2 + i(\sin y_1\cos y_2 + \cos y_1\sin y_2) \\ &= \cos(y_1+y_2) + i\sin(y_1+y_2) = e^{i(y_1+y_2)}. \end{aligned}$$

This verifies the formula.

By the formula, we have $e^{z_1} = e^{z_1-z_2}e^{z_2}$, which gives

$$\frac{e^{z_1}}{e^{z_2}} = e^{z_1-z_2},$$

and, as a special case, $\dfrac{1}{e^z} = e^{-z}$. Moreover, for any integer $k$,

$$(e^z)^k = e^{kz}.$$

### 1.4. **Derivative.**

The exponential function $e^z$ is entire and

$$(e^z)' = e^z, \quad z \in \mathbb{C},$$

because

$$\frac{\partial e^z}{\partial x} = e^z = -i\frac{\partial e^z}{\partial y}, \quad z \in \mathbb{C}.$$

**Example.** Solve the equation $e^z = 1 + i$.

Write the equation as

$$e^x e^{iy} = \sqrt{2}e^{i\pi/4}.$$

Then

$$e^x = \sqrt{2}, \quad y = \frac{\pi}{4} + 2n\pi, \quad n = 0, \pm1, \pm2, \ldots$$

We have

$$z = \frac{1}{2}\ln 2 + i\Big(\frac{\pi}{4} + 2n\pi\Big), \quad n = 0, \pm1, \pm2, \ldots$$

## 2. The Logarithmic Function

### 2.1. **Definition.**

For $z \neq 0$, the logarithmic function $\log z$ is defined as the multi-valued function,

$$\begin{aligned} \log z &= \ln|z| + i\arg z \\ &= \ln|z| + i\,(\mathrm{Arg} z + 2\pi n), \quad n = 0, \pm1, \pm2, \ldots\ . \end{aligned}$$

Define the principal value of $\log z$ to be

$$\mathrm{Log}\, z = \ln|z| + i\,\mathrm{Arg} z, \quad z \neq 0.$$

The values of $\log z$ are the complex numbers $w$ satisfying the equation $e^w = z$, that is,

$$e^{\log z} = z, \quad z \neq 0.$$

To see this, plugging

$$z = re^{i\Theta}, \quad -\pi < \Theta \le \pi, \quad w = u + iv,$$

into the equation $e^w = z$ gives

$$e^u e^{iv} = re^{i\Theta},$$

which implies that

$$e^u = r, \quad v = \Theta + 2n\pi,$$

that is,

$$w = \log r + i(\Theta + 2n\pi).$$

So the logarithmic function is the inverse function of the exponential function. However, for $z = x + iy$, we have

$$\log(e^z) = \ln|e^z| + i\arg(e^z) = \ln(e^x) + i(y + 2n\pi) = z + 2n\pi i, \quad n = 0, \pm 1, \pm 2, \dots .$$

**Example.**

(**1**) $\log 1 = \ln 1 + i2n\pi = 2n\pi i, \quad n = 0, \pm 1, \pm 2, \dots . \quad \operatorname{Log} 1 = 0.$

(**2**) $\log(-1) = \ln 1 + i(\pi + 2n\pi) = (2n+1)\pi i, \quad n = 0, \pm 1, \pm 2, \dots . \quad \operatorname{Log}(-1) = \pi i.$

(**3**) $\log(1+i) = \ln|1+i| + i\arg(1+i) = \ln\sqrt{2} + i\left(\frac{\pi}{4} + 2n\pi\right), \quad n = 0, \pm 1, \pm 2, \dots .$
$\operatorname{Log}(1+i) = \frac{1}{2}\ln 2 + \frac{i\pi}{4}.$

## 2.2. **Branches and Derivatives of Logarithms.**

Let $z = re^{i\theta}$. The logarithmic function is written

$$w = \log z = \ln r + i\theta,$$

where $\theta$ is multi-valued. One can restrict the range of $\theta$ so that $\log z$ is single valued. Let $\alpha \in \mathbb{R}$. Then the function

$$\log z = \ln r + i\theta, \quad r > 0, \ \alpha < \theta \le \alpha + 2\pi,$$

is single valued. But it is not continuous on the ray $\theta = \alpha + 2\pi$. Removing the ray gives the domain

$$D_\alpha = \{z \in \mathbb{C} : r > 0, \alpha < \theta < \alpha + 2\pi\}.$$

Then $\log z$ is continuous in $D_\alpha$. This is called a branch cut. We get a continuous branch of the logarithmic function,

$$\log z = \ln r + i\theta, \quad r > 0, \ \alpha < \theta < \alpha + 2\pi.$$

The principal branch is

$$\operatorname{Log} z = \ln r + i\theta, \quad r > 0, \ -\pi < \theta < \pi.$$

We have

$$(\log z)_r = \frac{1}{r}, \quad (\log z)_\theta = i.$$

which satisfy the Cauchy-Riemann equation,

$$(\log z)_r = \frac{1}{ir}(\log z)_\theta.$$

Therefore, $\log z$ is analytic in $D_\alpha$, and its derivative is

$$\frac{d}{dz}\log z = e^{-i\theta}(\log z)_r = \frac{1}{re^{i\theta}}.$$

That is,

$$\frac{d}{dz}\log z = \frac{1}{z}, \quad z \in D_\alpha,$$

and

$$\frac{d}{dz}\mathrm{Log}\, z = \frac{1}{z}, \quad z \in D_{-\pi}.$$

Formulas of the real logarithm are not always true for the complex logarithm.

**Example.** $\mathrm{Log}\,(i^3) \neq 3\,\mathrm{Log}\, i$.

We have

$$\mathrm{Log}\,(i^3) = \mathrm{Log}\,(-i) = -\frac{\pi}{2}i$$

and

$$3\,\mathrm{Log}\, i = 3\Big(i\frac{\pi}{2}\Big) = \frac{3\pi}{2}i.$$

## 2.3. **Addition Formula.**

The following addition formula is true,

$$\log(z_1 z_2) = \log z_1 + \log z_2, \quad z_1, z_2 \in \mathbb{C}.$$

This can be verified by the definition of $\log z$:

$$\log(z_1 z_2) = \log|z_1 z_2| + i\arg(z_1 z_2) = \log|z_1| + \log|z_2| + i(\arg z_1 + \arg z_2) = \log(z_1) + \log(z_2).$$

Note that the addition formula is for multi-valued functions, that is, the set of all values on the left-side of the equation is equal to the set of all values on the right-hand side of the equation. For a single branch, the formula may not be true, for example, one may have

$$\mathrm{Log}(z_1 z_2) \neq \mathrm{Log}(z_1) + \mathrm{Log}(z_2).$$

It can also be easily verified the subtraction formula,

$$\log\Big(\frac{z_1}{z_2}\Big) = \log z_1 - \log z_2.$$

**Example.** Let $z_1 = z_2 = -1$. We have

$$\log 1 = 2n\pi i, \quad \log(-1) = (2n+1)\pi i, \quad n = 0, \pm 1, \pm 2, \dots\ .$$

Since adding odd integers to odd integers gives even integers, we have

$$\log 1 = \log(-1) + \log(-1).$$

We also have

$$\mathrm{Log}\, 1 = 0, \quad \mathrm{Log}(-1) = \pi i,$$

and

$$\mathrm{Log}(z_1 z_2) = 0 \neq 2\pi i = \mathrm{Log}(-1) + \mathrm{Log}(-1).$$

By the addition and subtraction formulas and the identity $z = e^{\log z}$, we obtain the formula for integer powers,

$$z^n = e^{n \log z}, \quad z \neq 0, \ n = 0, \pm 1, \pm 2, \dots \ . \tag{2.1}$$

For roots, we have the formula,

$$z^{\frac{1}{n}} = \exp\Big(\frac{1}{n} \log z\Big), \quad n = 1, 2, \dots \ . \tag{2.2}$$

This can be seen because

$$\begin{aligned} \exp\Big(\frac{1}{n} \log z\Big) &= \exp\Big(\frac{1}{n} \ln r + \frac{i(\Theta + 2k\pi)}{n}\Big), \quad k = 0, \pm 1, \pm 2, \dots \\ &= r^{\frac{1}{n}} \exp\Big(\frac{i(\Theta + 2k\pi)}{n}\Big), \quad k = 0, 1, \dots, n-1 \\ &= z^{\frac{1}{n}}, \end{aligned}$$

where $z = r \exp(i\Theta)$, $\Theta = \operatorname{Arg} z$.

## 3. Power Functions

### 3.1. **Definition.**

Given a complex number $c$, the power function $z^c$, $z \neq 0$, is defined by

$$z^c = e^{c \log z}, \quad z \neq 0. \tag{3.1}$$

This definition is consistent with (2.1) and (2.2).

**Example.** (**1**) Find $i^{3i}$.

$$i^{3i} = \exp(3i \log i) = \exp\Big(3i\Big(i\frac{\pi}{2} + 2n\pi i\Big)\Big) = \exp\Big(-\frac{3\pi}{2} - 6n\pi\Big), \quad n = 0, \pm 1, \pm 2, \dots \ .$$

(**2**) Find $i^i$ and $i^{-i}$.

$$\begin{aligned} i^i &= \exp(i \log i) = \exp\Big(-\frac{\pi}{2} - 2n\pi\Big), \quad n = 0, \pm 1, \pm 2, \dots \ . \\ i^{-i} &= \exp(-i \log i) = \exp\Big(\frac{\pi}{2} + 2n\pi\Big), \quad n = 0, \pm 1, \pm 2, \dots \ . \end{aligned}$$

Then

$$i^i \, i^{-i} = \exp\big(2\pi n\big), \quad n = 0, \pm 1, \pm 2, \dots$$

and hence

$$i^i \, i^{-i} \neq i^0 = 1.$$

(**3**) Find $(\sqrt{3} + i)^{\frac{3}{2}}$.

$$(\sqrt{3} + i)^{\frac{3}{2}} = e^{\frac{3}{2}\log(\sqrt{3}+i)} = e^{\frac{3}{2}(\ln 2 + i(\frac{\pi}{6} + 2n\pi))} = 2\sqrt{2} e^{i(\frac{\pi}{4} + 3n\pi)} = \pm 2\sqrt{2} e^{i\frac{\pi}{4}} = \pm 2(1 + i).$$

**Reciprocal.** There is formula $\dfrac{1}{z^c} = z^{-c}$.

This follows from the property $\dfrac{1}{e^z} = e^{-z}$:

$$\frac{1}{z^c} = \frac{1}{e^{c\log z}} = e^{-c\log z} = z^{-c}.$$

### 3.2. **Derivative.**

Let $z = re^{i\theta}$ and $\alpha \in \mathbb{R}$, the branch of logarithm,

$$\log z = \ln r + i\theta, \quad r > 0,\ \alpha < \theta < \alpha + 2\pi,$$

is single-valued and analytic. This gives a branch of the power function,

$$z^c = e^{c\log z}, \quad z \in D_\alpha = \{z \in \mathbb{C} : |z| > 0, \alpha < \arg z < \alpha + 2\pi\},$$

which is single-valued and analytic. Its derivative can be calculated,

$$\frac{d}{dz}z^c = \frac{d}{dz}e^{c\log z} = \frac{c}{z}e^{c\log z} = c\frac{e^{c\log z}}{e^{\log z}} = ce^{(c-1)\log z},$$

Thus,

$$\frac{d}{dz}z^c = cz^{c-1}, \quad z \in D_\alpha.$$

### 3.3. **Principal Value.**

The principal value of $z^c$ is defined as

$$\text{P.V. } z^c = e^{c\,\mathrm{Log}\, z},$$

which is called the principal branch of $z^c$ in the domain $D_{-\pi}$.

Let $c_1, c_2$ and $z$, $z \neq 0$ be complex numbers. If all of the powers involved are principal values, then

$$z^{c_1+c_2} = z^{c_1}z^{c_2}, \quad \frac{z^{c_1}}{z^{c_2}} = z^{c_1-c_2}, \quad (z^c)^n = z^{cn}, \quad n = 1, 2, \dots\ .$$

**Example.** Let $z_1 = 1 + i$, $z_2 = 1 - i$, and $z_3 = -1 - i$. Then for principal values of powers,

$$\begin{aligned}
(z_1z_2)^i &= 2^i = e^{i\mathrm{Log}\,2} = e^{i\ln 2}\\
(z_2z_3)^i &= (-2)^i = e^{i\mathrm{Log}(-2)} = e^{i(\ln 2 + i\pi)} = e^{-\pi}e^{i\ln 2}\\
z_1^i &= e^{i\mathrm{Log}(1+i)} = e^{i(\ln\sqrt{2} + i\frac{\pi}{4})} = e^{-\frac{\pi}{4}}e^{\frac{i\ln 2}{2}}\\
z_2^i &= e^{i\mathrm{Log}(1-i)} = e^{i(\ln\sqrt{2} - i\frac{\pi}{4})} = e^{\frac{\pi}{4}}e^{\frac{i\ln 2}{2}}\\
z_3^i &= e^{i\mathrm{Log}(-1-i)} = e^{i(\ln\sqrt{2} - i\frac{3\pi}{4})} = e^{\frac{3\pi}{4}}e^{\frac{i\ln 2}{2}}
\end{aligned}$$

and

$$\begin{aligned}
(z_1z_2)^i &= z_1^iz_2^i\\
(z_2z_3)^i &= z_2^iz_3^ie^{-2\pi}.
\end{aligned}$$

Thus,

$$(z_2 z_3)^i \neq z_2^i z_3^i.$$

## 4. Trigonometric Functions

### 4.1. **Definition.**

Eucler's formula

$$e^{ix} = \cos x + i \sin x$$

implies that

$$\sin x = \frac{e^{ix} - e^{-ix}}{2i}, \quad \cos x = \frac{e^{ix} + e^{-ix}}{2}.$$

This leads to the definition of complex functions,

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i}, \quad \cos z = \frac{e^{iz} + e^{-iz}}{2}. \tag{4.1}$$

Euler's formula can be extended to the complex case.

$$e^{iz} = \cos z + i \sin z.$$

We define

$$\tan z = \frac{\sin z}{\cos z} = -i \frac{e^{iz} - e^{-iz}}{e^{iz} + e^{-iz}}. \tag{4.2}$$

**Example.** Find the roots of $\sin z = 1$.

The equation gives

$$e^{iz} - e^{-iz} = 2i, \quad \text{that is,} \quad e^{ix}e^{-y} - e^{-ix}e^{y} = 2i.$$

Then

$$\left(e^{y} - e^{-y}\right) \cos x = 0, \text{ and } \left(e^{y} + e^{-y}\right) \sin x = 2.$$

The solutions are $y = 0$, $x = \frac{\pi}{2} + 2n\pi$, that is, $z = \frac{\pi}{2} + 2n\pi$, $n = 0, \pm 1, \pm 2, \ldots$.

### 4.2. **Derivative.**

Since the exponential function is entire, both sine and cosine functions are entire. By

$$\frac{d}{dz} e^{iz} = i e^{iz}, \quad \frac{d}{dz} e^{-iz} = -i e^{-iz},$$

we have

$$\frac{d}{dz} \sin z = \cos z, \quad \frac{d}{dz} \cos z = -\sin z, \quad \frac{d}{dz} \tan z = \frac{1}{\cos^2 z}. \tag{4.3}$$

### 4.3. **Trigonometric Identities.**

By the definition (4.1), it is easily seen that

$$\sin(-z) = -\sin z, \quad \cos(-z) = \cos z.$$

By the definition (4.1) and the addition formula of exponential function, it can be shown that

$$\sin(z_1 + z_2) = \sin z_1 \cos z_2 + \cos z_1 \sin z_2,$$
$$\cos(z_1 + z_2) = \cos z_1 \cos z_2 - \sin z_1 \sin z_2.$$

By these identities, we get

$$\sin 2z = 2 \sin z \cos z, \quad \cos 2z = \cos^2 z - \sin^2 z,$$
$$\sin^2 z + \cos^2 z = 1,$$

and

$$\sin\left(z + \frac{\pi}{2}\right) = \cos z, \quad \cos\left(z + \frac{\pi}{2}\right) = -\sin z,$$
$$\sin(z + 2\pi) = \sin z, \quad \sin(z + \pi) = -\sin z,$$
$$\cos(z + 2\pi) = \cos z, \quad \cos(z + \pi) = -\cos z,$$
$$\tan(z + \pi) = \tan z, \quad \tan\left(z + \frac{\pi}{2}\right) = -\frac{1}{\tan z}.$$

## 5. Hyperbolic Functions

### 5.1. **Definition and Identities.**

The hyperbolic sine and hyperbolic cosine functions are defined by

$$\sinh z = \frac{e^z - e^{-z}}{2}, \quad \cosh z = \frac{e^z + e^{-z}}{2}. \tag{5.1}$$

Both functions are entire. Their derivatives are

$$\frac{d}{dz} \sinh z = \cosh z, \quad \frac{d}{dz} \cosh z = \sinh z.$$

There are hyperbolic identities,

$$\sinh(-z) = -\sinh z, \quad \cosh(-z) = \cosh z,$$
$$\cosh^2 z - \sinh^2 z = 1,$$
$$\sinh(z + 2\pi i) = \sinh z, \quad \cosh(z + 2\pi i) = \cosh z,$$
$$\sinh(z_1 + z_2) = \sinh z_1 \cosh z_2 + \cosh z_1 \sinh z_2,$$
$$\cosh(z_1 + z_2) = \cosh z_1 \cosh z_2 + \sinh z_1 \sinh z_2.$$

### 5.2. **Zeros of Trigonometric and Hyperbolic Functions.**

The trigonometric and hyperbolic functions have the relations:

$$\cosh(iz) = \cos z, \quad \cos(iz) = \cosh z,$$
$$\sinh(iz) = i \sin z, \quad \sin(iz) = i \sinh z. \tag{5.2}$$

By these identities and

$$\sin(x+iy) = \sin x \cos(iy) + \cos x \sin(iy),$$

we obtain

$$\sin z = \sin x \cosh y + i \cos x \sinh y, \quad z = x + iy.$$

It gives

$$|\sin z|^2 = \sin^2 x \cosh^2 y + \cos^x \sinh^2 y,$$

and hence

$$|\sin z|^2 = \sin^2 x + \sinh^2 y, \quad z = x + iy. \tag{5.3}$$

This implies that $\sin z = 0$ if and only if $\sin x = 0$ and $\sinh y = 0$, that is, $x = n\pi$, $n = 0, \pm 1, \pm 2, \ldots$, and $y = 0$. Therefore, the zeros of $\sin z$ are given by

$$\sin z = 0 \text{ if and only if } z = n\pi, \ n = 0, \pm 1, \pm 2, \ldots . \tag{5.4}$$

Since

$$\cos z = \cos(-z) = \sin\Big(-z + \frac{\pi}{2}\Big) = -\sin\Big(z - \frac{\pi}{2}\Big),$$

the zeros of $\cos z$ are given by

$$\cos z = 0 \text{ if and only if } z = \frac{\pi}{2} + n\pi, \ n = 0, \pm 1, \pm 2, \ldots . \tag{5.5}$$

By (5.2),

$$|\sinh z| = |\sin(iz)|, \quad \text{that is, } |\sinh z| = |\sin(-y + ix)|.$$

By this and (5.3), we get

$$|\sinh z|^2 = \sinh^2 x + \sin^2 y, \quad z = x + iy. \tag{5.6}$$

This implies that the zeros of $\sinh z$ are given by

$$\sinh z = 0 \text{ if and only if } z = n\pi i, \ n = 0, \pm 1, \pm 2, \ldots . \tag{5.7}$$

By (5.2) and (5.5), the zeros of $\cosh z$ are given by

$$\cosh z = 0 \text{ if and only if } z = \Big(\frac{\pi}{2} + n\pi\Big)i, \ n = 0, \pm 1, \pm 2, \ldots . \tag{5.8}$$

**Example.** Find the roots of the equation $\sinh z = 2i$.

Since $\sinh(iz) = i \sin z$, we solve the equation $\sin z = 2$. We have

$$\sin z = \sin x \cosh y + i \cos x \sinh y = 2.$$

This gives

$$\sin x \cosh y = 2, \quad \cos x \sinh y = 0.$$

Then

$$\cos x = 0, \quad \sin x = 1, \quad \cosh y = 2.$$

The solutions are

$$x = \frac{\pi}{2} + 2n\pi, \quad y = \ln(2 \pm \sqrt{3}), \quad n = 0, \pm 1, \pm 2, \ldots .$$

Thus, $\sinh(iz) = 2i$ has roots, $z = \frac{\pi}{2} + 2n\pi + i\ln(2 \pm \sqrt{3})$. The equation $\sinh z = 2i$ has roots,

$$z = i(\frac{\pi}{2} + 2n\pi) - \ln(2 \pm \sqrt{3}) = \pm\ln(2 + \sqrt{3}) + i(\frac{\pi}{2} + 2n\pi), \quad n = 0, \pm 1, \pm 2, \ldots.$$

## 6. Inverse Trigonometric and Hyperbolic Functions

The inverse function $\sin^{-1} z$ is defined by

$$w = \sin^{-1} z \quad \text{if} \quad z = \sin w.$$

The equation $z = \sin w$ can be written as

$$z = \frac{e^{iw} - e^{-iw}}{2i}, \quad \text{i.e.} \quad (e^{iw})^2 - 2ize^{iw} - 1 = 0.$$

The solution is

$$e^{iw} = iz + (1 - z^2)^{\frac{1}{2}},$$

and hence

$$w = \sin^{-1} z = -i\log(iz + (1 - z^2)^{\frac{1}{2}}).$$

This inverse function is a multi-valued function.

**Example.**

$$\begin{aligned}
\sin^{-1}(-i) &= -i\log(1 \pm \sqrt{2}) \\
&= \begin{cases} -i\log(1 + \sqrt{2}) = -i\ln(1 + \sqrt{2}) + 2n\pi & n = 0, \pm 1, \pm 2, \ldots \\ -i\log(1 - \sqrt{2}) = -i\ln(\sqrt{2} - 1) + (2n + 1)\pi & n = 0, \pm 1, \pm 2, \ldots \end{cases} \\
&= n\pi + i(-1)^{n+1}\ln(1 + \sqrt{2}), \quad n = 0, \pm 1, \pm 2, \ldots \ .
\end{aligned}$$

Similarly, we have

$$\begin{aligned}
\cos^{-1} z &= -i\log(z + i(1 - z^2)^{\frac{1}{2}}) \\
\tan^{-1} z &= \frac{i}{2}\log\frac{i + z}{i - z} \\
\sinh^{-1} z &= \log(z + (z^2 + 1)^{\frac{1}{2}}) \\
\cosh^{-1} z &= \log(z + (z^2 - 1)^{\frac{1}{2}}) \\
\tanh^{-1} z &= \frac{1}{2}\log\frac{1 + z}{1 - z}.
\end{aligned}$$

There are formulas for derivatives,

$$\begin{aligned}
\frac{d}{dz}\sin^{-1} z &= \frac{1}{(1 - z^2)^{\frac{1}{2}}} \\
\frac{d}{dz}\cos^{-1} z &= \frac{-1}{(1 - z^2)^{\frac{1}{2}}} \\
\frac{d}{dz}\tan^{-1} z &= \frac{1}{1 + z^2}.
\end{aligned}$$

## References

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